

Chapter 2

Structure of a Java Program

Case sensitive Language- Java different b/w upper case and lower case

1. Helloworld.Java

```
public class HelloWorld {  
  
    public static void main(String[] args) {  
  
        System.out.println("Hello, World!");  
  
    }  
  
}
```

Use of Compiler and Interpreter in Java

- public class HelloWorld → Every Java program must have at least one class.
- The class name must match the filename
- public static void main(String[] args) → The main method is the entry point of any Java program.
- System.out.println(""); → Prints text to the screen

SYNTAX RULE

- ✓ Every statement ends with a semicolon (;).
- ✓ Curly braces {} define blocks of code.
- ✓ Class names should start with an uppercase letter.
- ✓ Method and variable names should start with a lowercase letter.
- ✓ Java is case-sensitive: Hello ≠ hello.
- ✓ The file name must match the public class name.

DATA TYPES

What kind of variables can store and how much it will take

int age - 18;

double price - 99.50;

String name - "John";

Int, double, string—> data types

Types of data types

1. Primitive data types —> fixed size & specific purpose

Data Type	Size (in bytes)	Bits	Example Value	Description
byte	1 byte	8 bits	100	Small integer (-128 to 127)
short	2 bytes	16 bits	20000	Medium integer (-32,768 to 32,767)
int	4 bytes	32 bits	50000	Standard integer (most used)
long	8 bytes	64 bits	123456789L	Large integer
float	4 bytes	32 bits	5.75f	Decimal (single precision)
double	8 bytes	64 bits	19.99	Decimal (double precision)
char	2 bytes	16 bits	'A'	Single character (Unicode)
boolean	1 bit*	—	↓ true / false	Logical values

Program Example

```
public class DataTypeExample {  
    public static void main(String[] args) {  
        byte age = 25;  
        short year = 2025;  
        int salary = 50000;  
        long population = 80000000000L;  
        float height = 5.9f;  
        double weight = 65.8;  
        char grade = 'A';  
        boolean isPassed = true;  
    }  
}
```

```
        System.out.println("Age: " + age);  
        System.out.println("Year: " + year);  
        System.out.println("Salary: " + salary);  
        System.out.println("Population: " +  
        population);  
        System.out.println("Height: " + height);  
        System.out.println("Weight: " + weight);  
        System.out.println("Grade: " + grade);  
        System.out.println("Passed: " + isPassed);  
    }  
}
```

2. Non primitive data type —> program which is created by object, doesn't store value directly and store as address

Data Type	Description	Example
String	Sequence of characters (text)	<code>String name = "Alice";</code>
Array	Collection of similar data types	<code>int[] marks = {90, 85, 88};</code>
Class	Blueprint for creating objects	<code>class Student { ... }</code>
Object	Instance of a class	<code>Student s1 = new Student();</code>
Interface	Contract that a class must follow	<code>interface Shape { ... }</code>

Program Example for both data types

```
public class Example {  
    public static void main(String[] args) {  
        int num = 10;           // primitive  
        String name = "Java";   // non-primitive  
  
        System.out.println(num + 5); // arithmetic operation  
        System.out.println(name.toUpperCase()); // method on String  
    }  
}
```